Edgar Fong

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Skills

- Python | C++ | Java | JavaScript | C | C# | Linux | Oracle DB | MySQL | MongoDB | Terraform | React | Selenium | jQuery | MATHLAB | Git
- Azure | AWS | GCloud | Cloud Computing | CI/CD | Keras | TensorFlow | Pytorch | IoT | OOP | Unity 2D | Game Development
- Agile | Scrum | Full-Stack | Frontend | Backend | Programmable Logic Controller | English, Spanish – *All professional proficiency or above*

Professional Experience

Field Engineer | Adcom Wireless | **06/2019 - Current**

- Collaborated with major telecommunications companies (AT&T, Verizon, T-Mobile, Dish) to provide technical support and maintenance for telecommunications equipment, including antennas and radios, ensuring up time for millions of customers.
- Assisted in designing site layouts and equipment positions for optimal signal strength and performance across hundreds of sites.
- Configured and installed Programmable Logic Controllers (PLCs) and associated components, ensuring full operational status of cellular towers.
- Integrated new telecommunications sites during off-peak hours, reducing potential service disruptions by 80%.
- Communicated closely with clients to understand and fulfill technical requirements, ensuring high levels of customer satisfaction and adherence to project timelines.

Projects

Automated Cloud Instance Benchmarking Tool and Analysis

- Designed and implemented a cross-platform cloud performance testing tool in C#, automating VM provisioning and benchmarking across AWS, Azure, and GCP, with full compatibility on both Linux and Windows, reducing manual setup time by approximately 60%.
- Engineered a secure CLI application to streamline user setup, encrypt credentials in .env files, and run parallel, custom performance tests.
- Built a Node.js and Express.js web application hosted on Azure; visualizing hundreds of normalized performance test results stored in a MongoDB database hosted on AWS, enabling cost-performance analysis through dynamic, interactive dashboards.
- Spearheaded performance benchmarking workflows, executing tailored tests to assess CPU, memory, and disk performance.

Deep Learning Autonomous Tennis Ball Collector

- Created and optimized deep learning model through quantization to run efficiently on Raspberry Pi's ARM architecture, enabling real-time inference for autonomous operation.
- Conducted extensive data collection and model training using Keras and TensorFlow Lite, capturing and processing over 4000 images to enhance the deep learning model's performance in varied environmental conditions.
- Developed and applied computer vision algorithms using Python and OpenCV to enable real-time detection and precise localization of tennis balls, achieving a detection accuracy of approximately 83%.
- Led the design and assembly of hardware components including Raspberry Pi 4, TT motors, and custom scooper mechanism.
- Implemented PID control algorithms to optimize motor performance and ensure precise navigation and maneuverability of the autonomous tennis ball collector.

OCR Mobile App Data Scraper

- Developed an autonomous data scraping application using Python, automating the process of data collection from mobile apps.
- Utilized PyAutoGUI for screen navigation and data location, and applied image enhancement techniques for improved OCR recognition.
- Integrated the application with a database, ensuring efficient data upload and management with over 50,000 unique entries uploaded.
- Implemented multiple automation libraries (win32gui, keyboard, time) to achieve full automation of the data scraping process.

Static and Dynamic Malware and Botnet Analysis

- Performed static and dynamic analysis to understand encryption methods and detection techniques.
- Executed a controlled live demo showcasing the creation and deployment of a botnet and ransomware, emphasizing the security risks and educational aspects.
- Documented analysis results, highlighting vulnerabilities and mitigation strategies as well as safe coding practices.

Unity 2D Game Development

- Created a 2D game in Unity utilizing many of Unity's core features like C# and Unity's scriptable objects.
- Developed and maintained scripts for game mechanics, including player/enemy behavior, collectibles, health, and weapon systems.
- Enhanced game performance by implementing object pooling, particle systems, and custom animations, reducing memory usage by 30%.

Education

Bachelor of Science | California State University San Marcos | *San Marcos, CA, USA* | **08/2022 - 08/2024**

- Major in Computer Science

Associate of Science | Southwestern College | *San Diego, CA, USA* | **07/2019 - 05/2022**

- Associate in Computer Science